

Markscheme

Mathematics: analysis and approaches
Practice question bank

131. (a)

Given that $z^2 = 5 - 12i$, where $z = a + bi$ and a, b are real.

Find the value of a and the value of b for the root with $a > 0$.

equate real and imaginary parts of $(a + bi)^2$: $a^2 - b^2 = 5$ and $2ab = -12$ (M1)

substitute $b = -\frac{6}{a}$: $a^4 - 5a^2 - 36 = 0$ (M1) A1

solve for a^2 : $(a^2 - 9)(a^2 + 4) = 0 \Rightarrow a^2 = 9$ (M1)

since $a > 0$: $a = 3$, $b = -2$ A1

[6 marks]